

Inverse Problems in Geophysics

Part 9: Newtons method

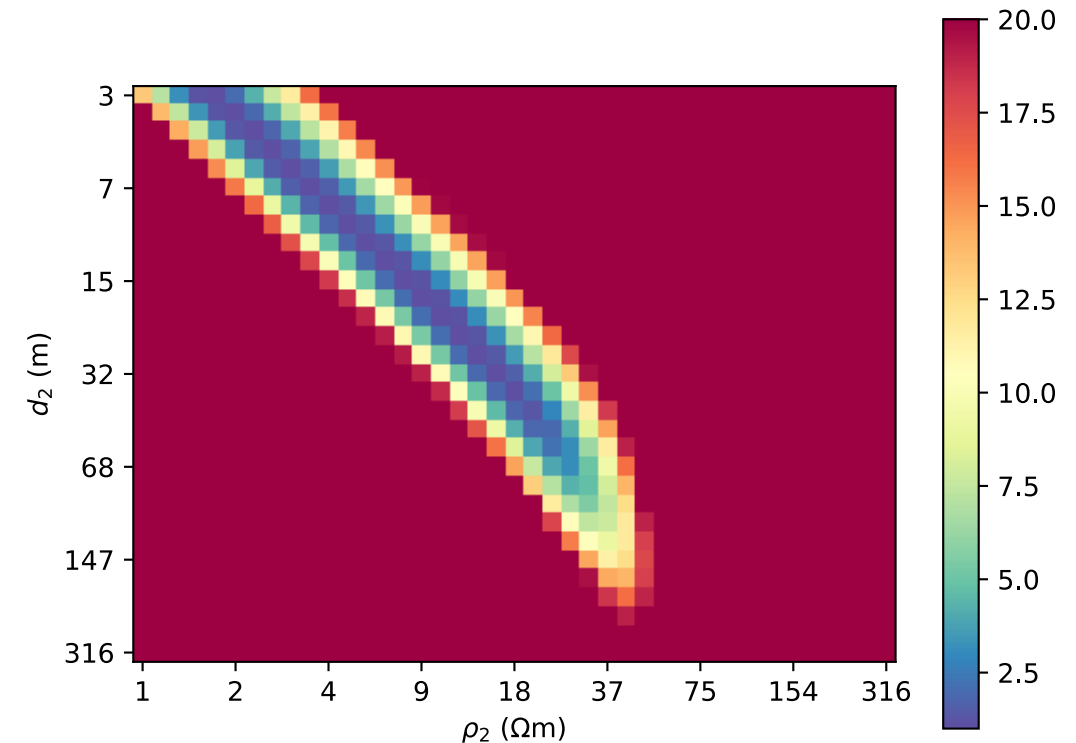
2. MGPY+MGIN

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Recap

- linear problems: least-squares solution of (regularized) problem
- non-linear problems: linearization of $f(m) \Rightarrow$ linear problem for $\Delta \mathbf{m}$ and $\Delta \mathbf{d} = \mathbf{d} - \mathbf{f}(\mathbf{m})$
- **grid search**: systematic search through model space



Objective function for VES

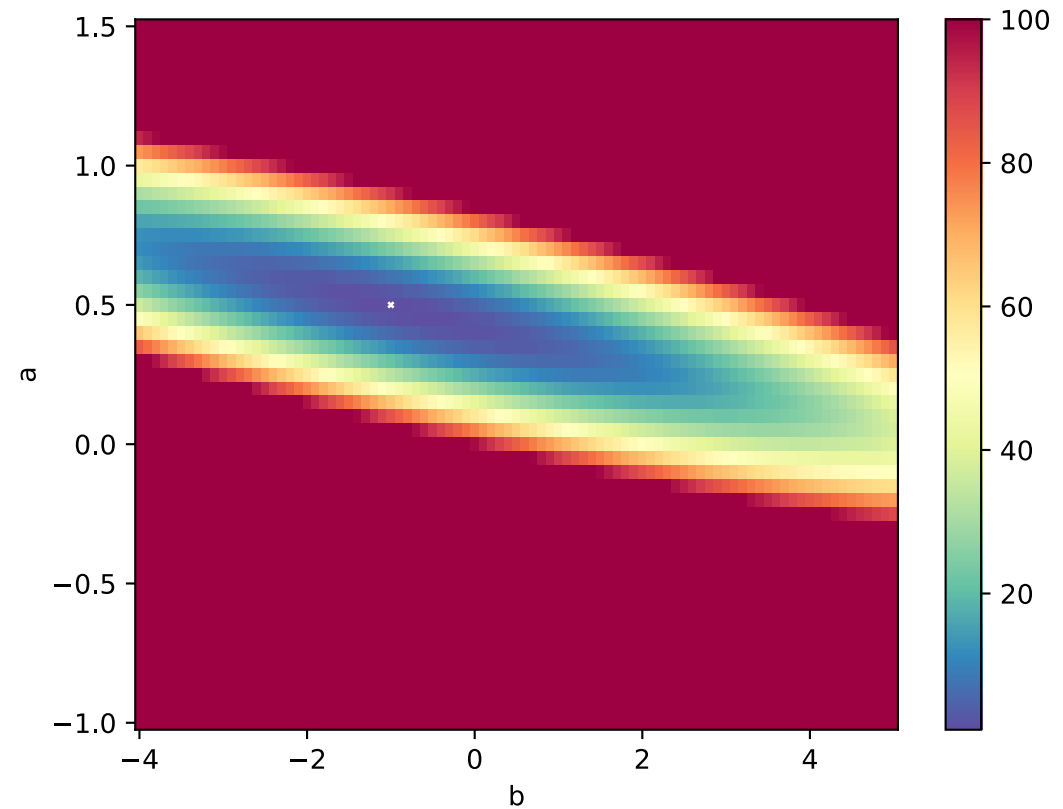
Objective function for linear problems

$$\Phi_d = (\mathbf{d} - \mathbf{G}\mathbf{m})^T (\mathbf{d} - \mathbf{G}\mathbf{m}) =$$

$$(\mathbf{d}^T - \mathbf{m}^T \mathbf{G}^T)(\mathbf{d} - \mathbf{G}\mathbf{m}) =$$

$$\mathbf{d}^T \mathbf{d} + \mathbf{m}^T \mathbf{G}^T \mathbf{G} \mathbf{m} - 2\mathbf{d}^T \mathbf{G} \mathbf{m}$$

$\Rightarrow$ a parabola for all m_i



Regularized non-linear problems

Problem

Regularization of linearized problem $\mathbf{S}\Delta\mathbf{m} - \mathbf{d}$

Regularization is applied to $\Delta\mathbf{m}$?

Good for truncation, damping, but not good for smoothing

$\Rightarrow$ adding smoothed $\Delta\mathbf{m}$ can lead to strange $\mathbf{m}$

rigorous minimization scheme required (today's lecture)

but: damping for $\Delta\mathbf{m}$ might be better than damping $\mathbf{m}$

Reference-based (linear) inversion

Linear problem: $\mathbf{f}(\mathbf{m}) = \mathbf{G}\mathbf{m}$: $\frac{\partial f_i}{\partial m_j} = G_{ij}$

plus regularization: $\|\mathbf{d} - \mathbf{G}\mathbf{m}\|^2 + \lambda\|\mathbf{C}\mathbf{m}\|^2 \rightarrow \min$

use $\mathbf{m}^0$ as a reference ($\mathbf{m} = \mathbf{m}^0 + \Delta\mathbf{m}$) with $\Delta\mathbf{d} = \mathbf{d} - \mathbf{G}\mathbf{m}^0$:

plus regularization: $\|\Delta\mathbf{d} - \mathbf{G}\Delta\mathbf{m}\|^2 + \lambda\|\mathbf{C}\Delta\mathbf{m}\|^2 \rightarrow \min$

e.g., $\mathbf{C} = \mathbf{I}$, keep the difference to $\mathbf{m}^0$ small and not $\mathbf{m}$

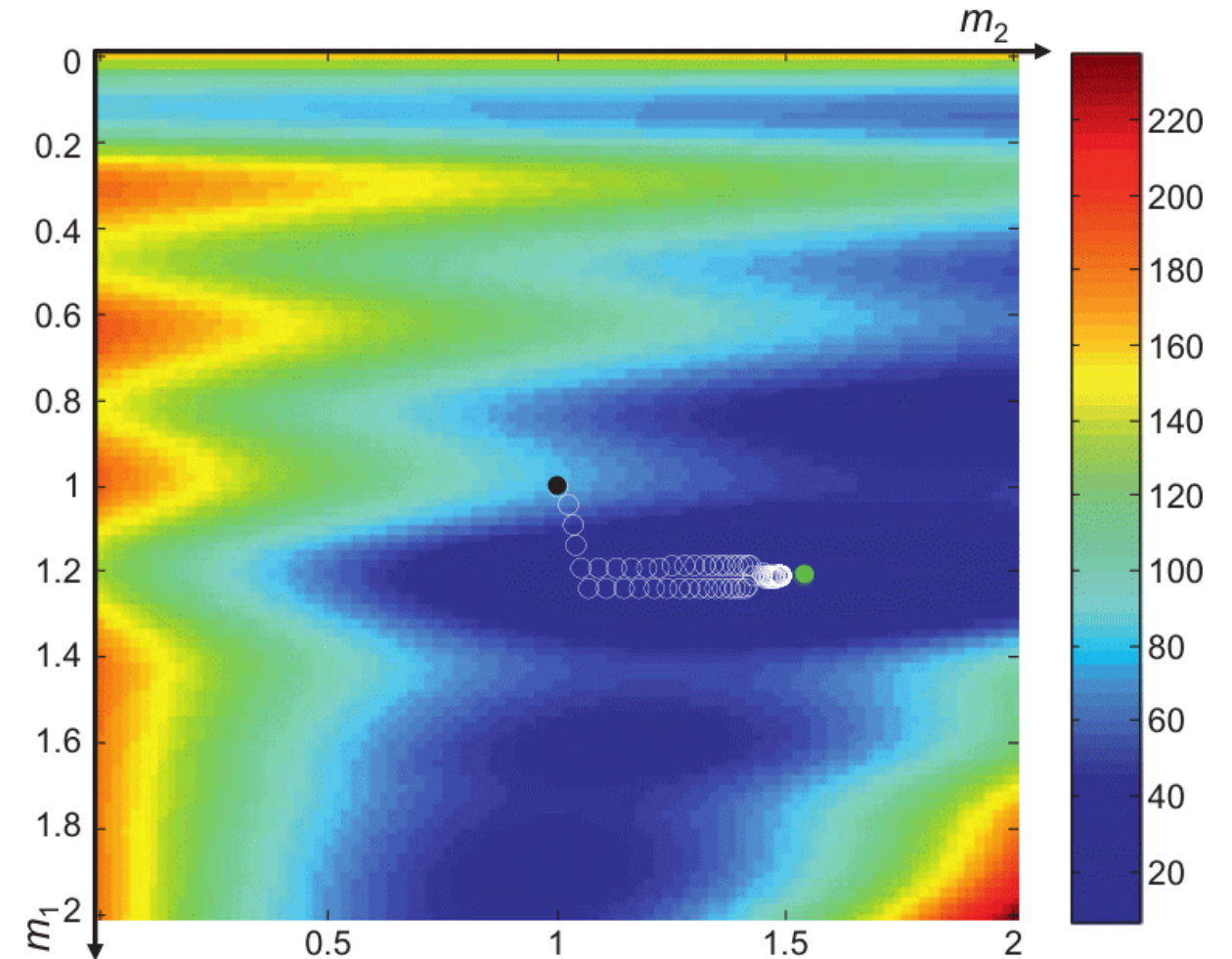
Gradient-based inversion

To minimize Φ_d , it makes sense to go down its gradient $\partial\Phi_d/\partial m_i$

i Steepest descent method

Go into the direction where Φ_d decreases:

$$\nu = -\frac{\nabla\Phi_d}{|\nabla\Phi_d|}$$



Gradient method

Line search using Armijo rule

$$\mathbf{m}^{k+1} = \mathbf{m}^k + \alpha \nu$$

i Armijo's rule to determine α

$$\Phi_d(\mathbf{m}^{k+1}) \leq \Phi_d(\mathbf{m}^k) + c\alpha \nu^T \nabla \Phi_d$$

- c -empirical constant, typically 1e-4
- start with large α and decrease until rule is fulfilled

$\Rightarrow$ requires a lot of forward computations

Newton's method

Taylor series expansion for the objective function:

$$\begin{aligned}\Phi_d(\mathbf{m}) = & \Phi_d(\mathbf{m}^0) + \sum_i^M b_i(m_i - m_i^0) + \\ & + \frac{1}{2} \sum_i^M \sum_j^M B_{ij}(m_i - m_i^0)(m_j - m_j^0) + \dots\end{aligned}$$

with **gradient vector** $b_i = \frac{\partial \Phi_d}{\partial m_i}$ and **Hessian matrix** $B_{ij} = \frac{\partial^2 \Phi_d}{\partial m_i \partial m_j}$

Newton's method

$$\dots \sum_i^M b_i(m_i - m_i^0) + \frac{1}{2} \sum_i^M \sum_j^M B_{ij}(m_i - m_i^0)(m_j - m_j^0)$$

Minimum: derivative zero

$$\frac{\partial \Phi}{\partial m_k} = 0 = b_k + \sum_j B_{kj}(m_k - m_k^0) \Rightarrow \mathbf{m} - \mathbf{m}^0 = -\mathbf{B}^{-1} \mathbf{b}$$

$$\Rightarrow \mathbf{m} = \mathbf{m}^0 - \mathbf{B}^{-1} \mathbf{b}$$

Newton's method for linear problems

$$\Phi_d = (\mathbf{d} - \mathbf{G}\mathbf{m})^T (\mathbf{d} - \mathbf{G}\mathbf{m})$$

$$b_i = \frac{\partial \Phi_d}{\partial m_i} = -2\mathbf{G}^T (\mathbf{d} - \mathbf{G}\mathbf{m})$$

$$B_{ij} = \frac{\partial^2 \Phi_d}{\partial m_i \partial m_j} = 2\mathbf{G}^T \mathbf{G}$$

$$\Rightarrow \mathbf{m} = -\mathbf{B}^{-1} \mathbf{b} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \mathbf{d} \text{ (least-squares solution)}$$

Non-linear: Gauss-Newton method

$$\Phi_d = (\mathbf{d} - \mathbf{f}(\mathbf{m}))^T (\mathbf{d} - \mathbf{f}(\mathbf{m}))$$

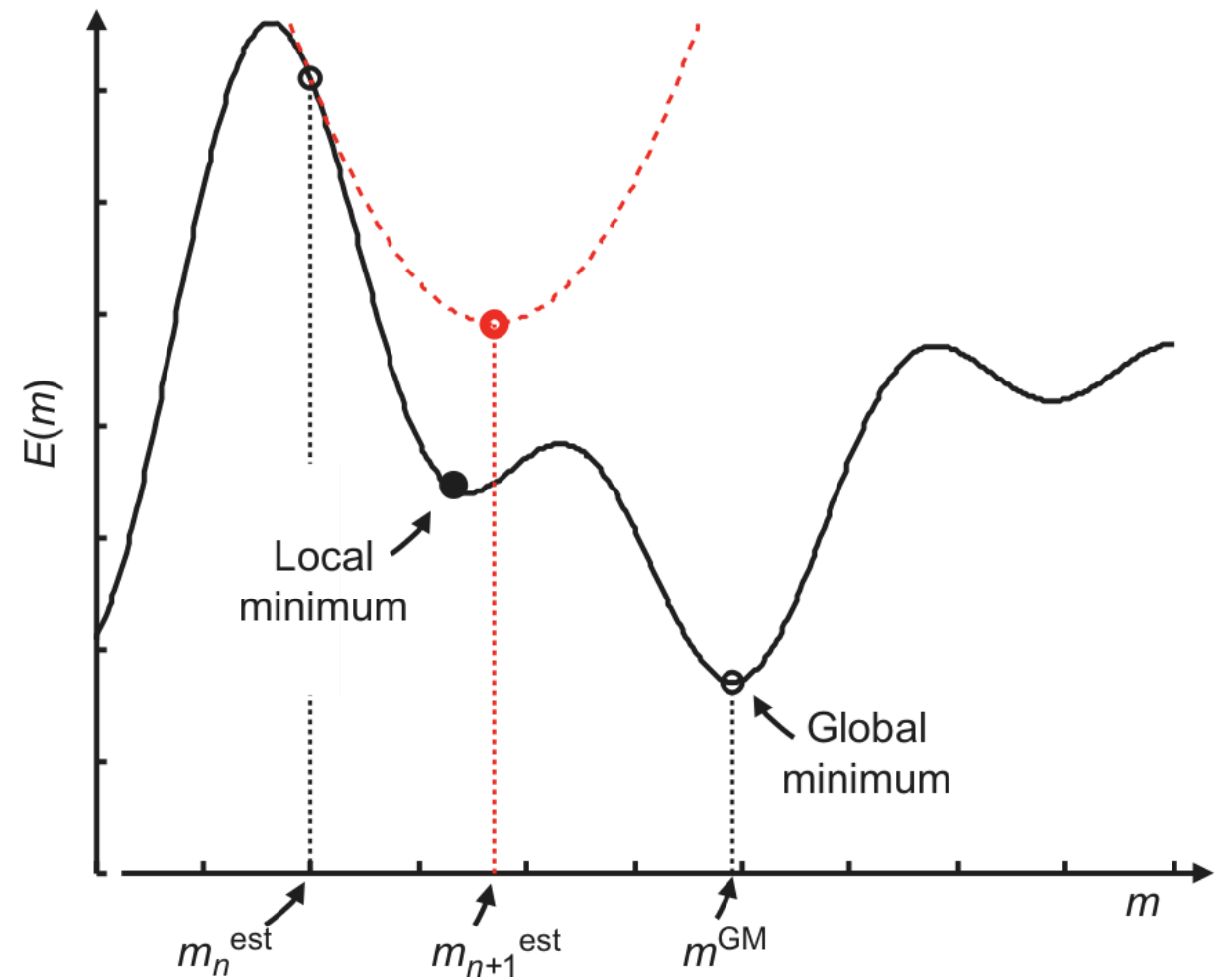
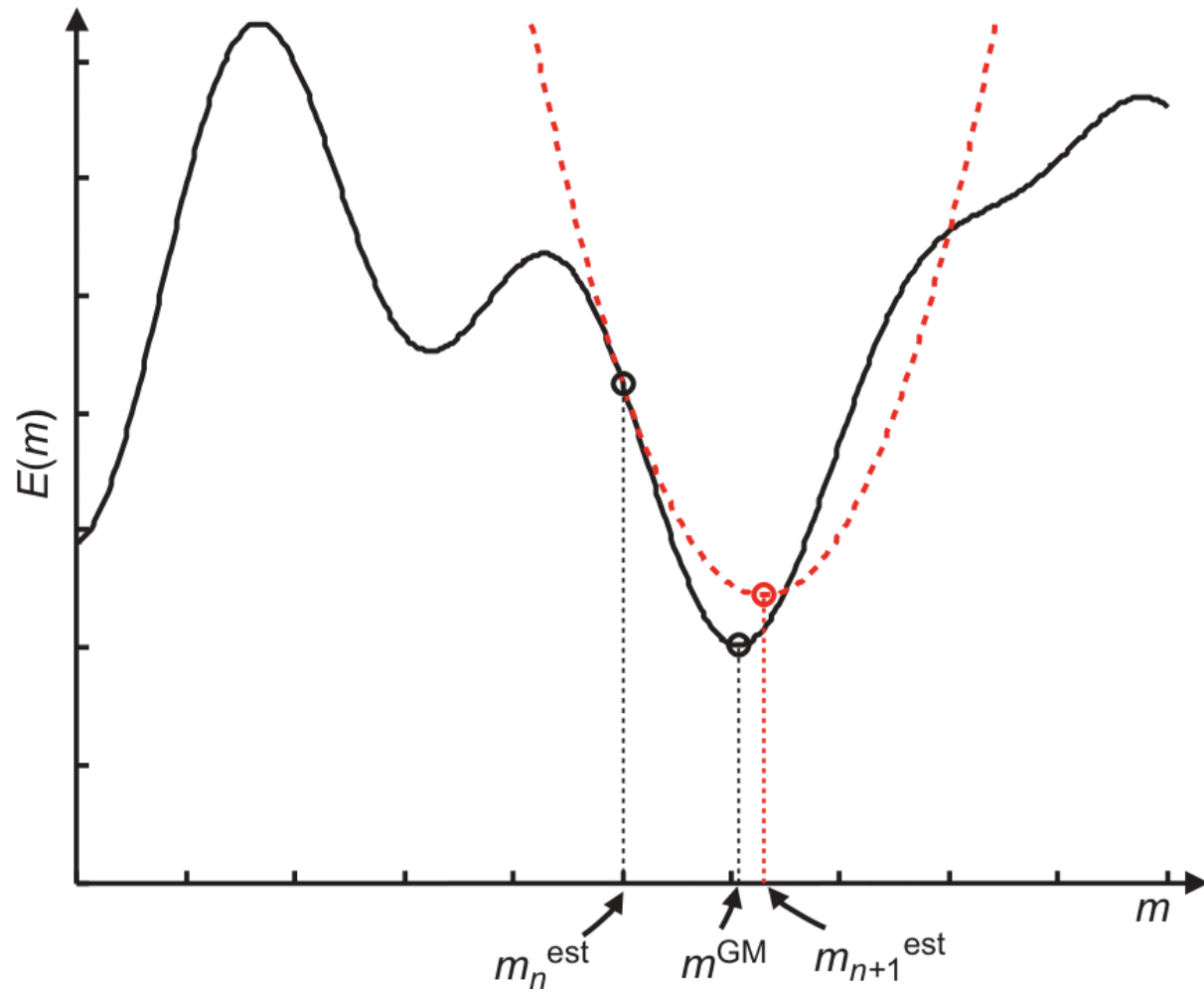
$$b_i = \frac{\partial \Phi_d}{\partial m_i} = -2\mathbf{S}^T (\mathbf{d} - \mathbf{f}(\mathbf{m})) \quad \text{with} \quad S_{ij} = \frac{\partial f_i(\mathbf{m}^0)}{\partial m_j}$$

$$B_{ij} = \frac{\partial^2 \Phi_d}{\partial m_i \partial m_j} = \frac{\partial b_i}{\partial m_j} \approx 2\mathbf{S}^T \mathbf{S}$$

$$\Rightarrow \mathbf{m} - \mathbf{m}^0 = -\mathbf{B}^{-1} \mathbf{b} = (\mathbf{S}^T \mathbf{S})^{-1} \mathbf{S}^T (\mathbf{d} - \mathbf{f}(\mathbf{m}))$$

$\Rightarrow$ least-squares solution for $\Delta \mathbf{m} / \Delta \mathbf{d}$

Linearization (Menke, 2012)



Gauss-Newton minimization

1. Choose starting model $\mathbf{m}^0$ and set $n=0$
2. Compute model response $\mathbf{f}(\mathbf{m}^n)$
3. Compute sensitivity matrix $\mathbf{S}^n$
4. Solve linearized subproblem $\mathbf{S}^n \Delta \mathbf{m}^n = \Delta \mathbf{d} = (\mathbf{d} - \mathbf{f}(\mathbf{m}^n))$
5. Update model by $\mathbf{m}^{n+1} = \mathbf{m}^n + \Delta \mathbf{m}^n$
6. If convergence quit, otherwise $n \leftarrow n + 1$ & proceed with 2.

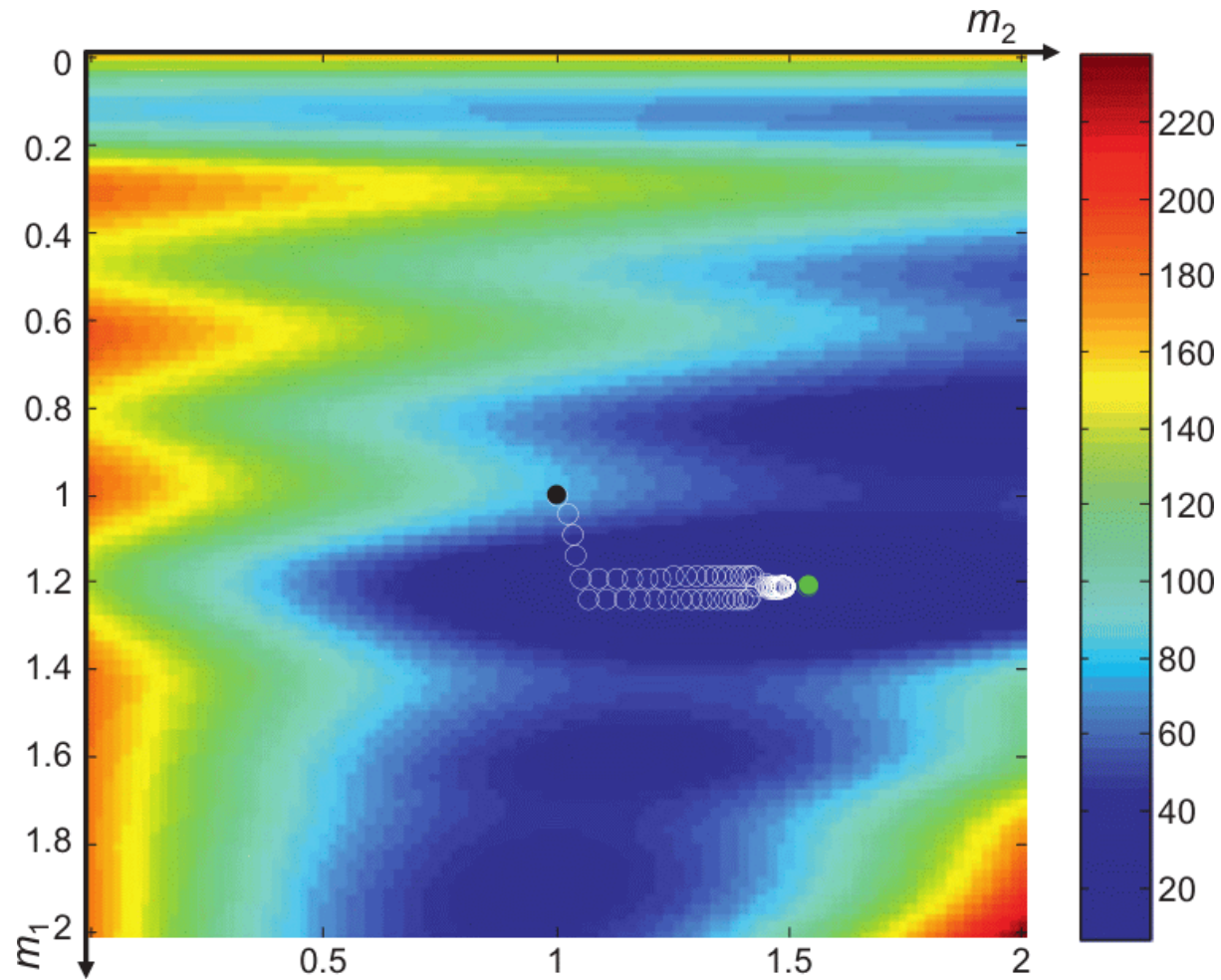
Line search for strong non-linearity

The Taylor approximation might change along $\Delta \mathbf{m}$, so we optimize the step length τ by “searching along the line”

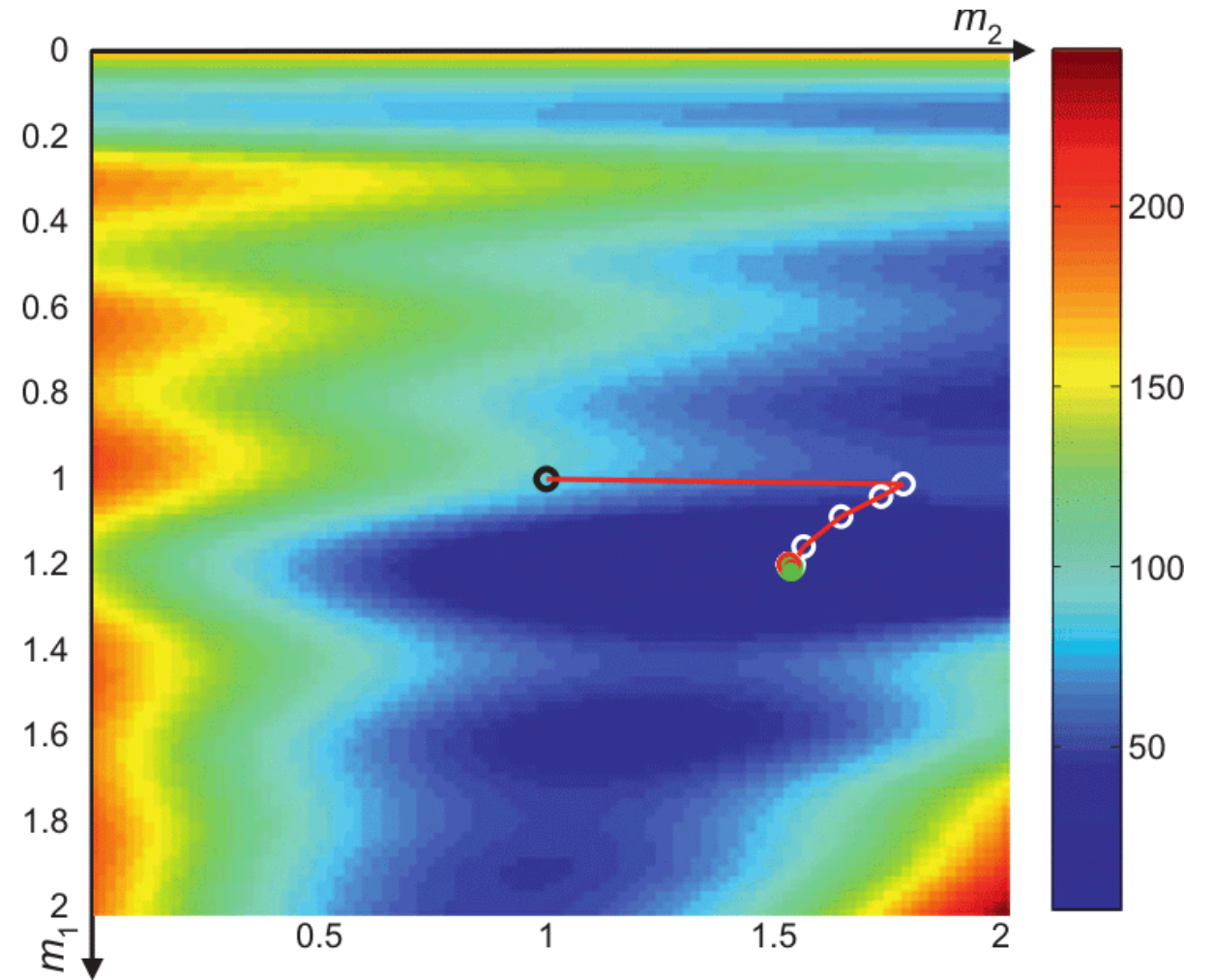
$$\mathbf{m}^{n+1} = \mathbf{m}^n + \tau^n \Delta \mathbf{m}^n$$

- exact line search: testing $\mathbf{f}(\mathbf{m} + \tau \Delta \mathbf{m})$ for many τ
- interpolation of $\mathbf{f}(\mathbf{m}^n + \tau \Delta \mathbf{m})$ between $\mathbf{f}(\mathbf{m}^n)$ & $\mathbf{f}(\mathbf{m}^n + \Delta \mathbf{m})$
- fit parabola through points $\mathbf{f}(\mathbf{m}^n)$, $\mathbf{f}(\mathbf{m}^n + \Delta \mathbf{m}/2)$, $\mathbf{f}(\mathbf{m}^n + \Delta \mathbf{m})$

Gradient vs. Newton



Gradient method



Newtons method

Computation of the sensitivity matrix

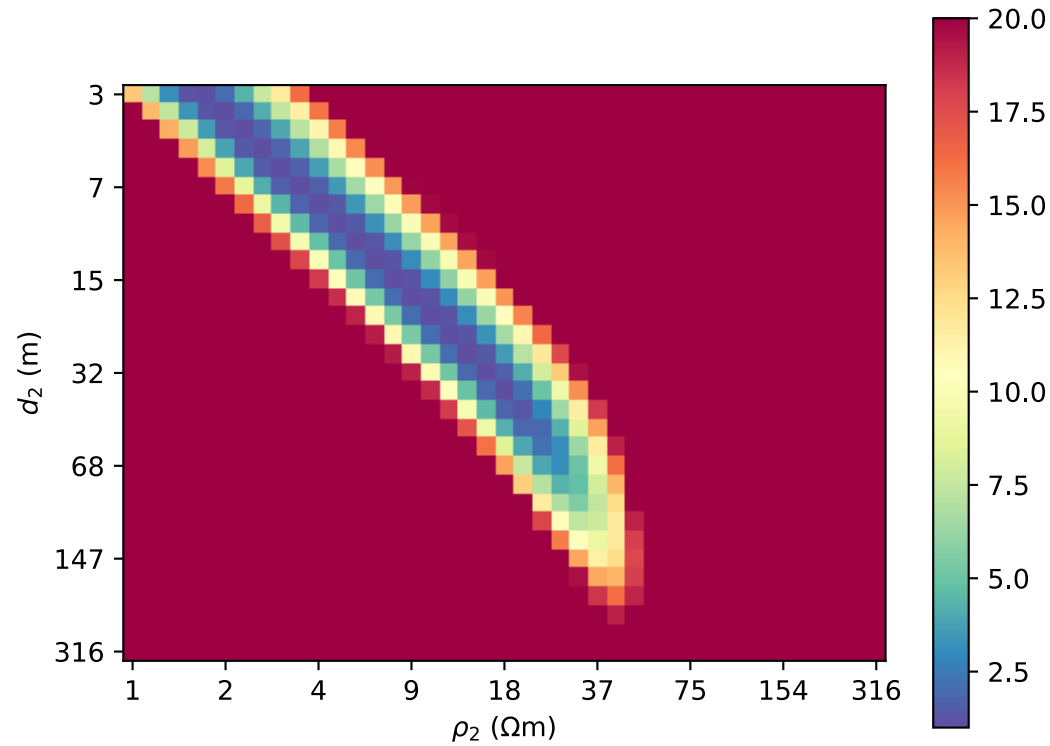
- analytically (derivation of the forward operator)
- transforming the PDE and its numerical solution
- perturbation method (brute force)

$$\frac{\partial f_i(\mathbf{m})}{\partial m_j} \approx \frac{f_i(\mathbf{m} + \delta_j \Delta m) - f_i(\mathbf{m})}{\Delta m}$$

with the Dirac vector $\delta_j = [0, \dots, 0, 1, 0, \dots, 0]^T$

$\Rightarrow$ one full forward computation for every model parameter

Model transformation



Objective function for VES

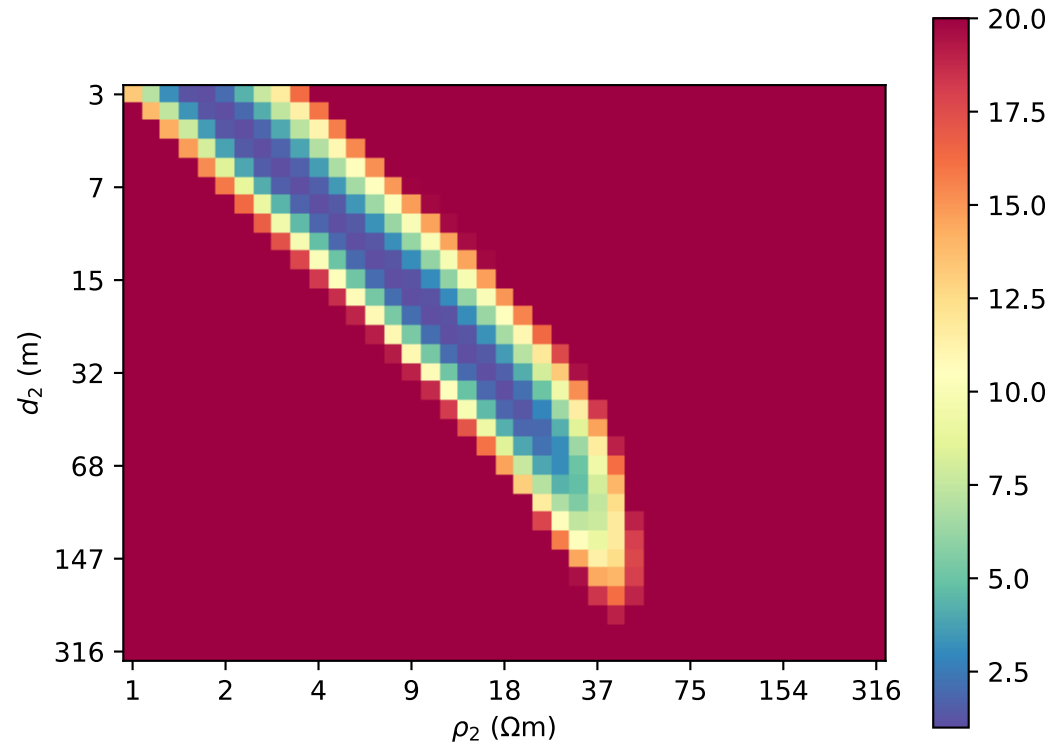
To keep the parameters positive, we often invert for the logarithms.

If we invert for $\hat{m}$ instead of m , we use the chain rule

$$\frac{\partial f}{\partial \hat{m}} = \frac{\partial f}{\partial m} \cdot \frac{\partial m}{\partial \hat{m}} = \frac{\partial f}{\partial m} / \frac{\partial \hat{m}}{\partial m}$$

$$\text{E.g. } \partial \log \rho / \partial \rho = 1/\rho$$

Data transformation



Objective function for VES

Often, measured data show a wide range so that we use the logarithm

If we invert $\hat{d}$ instead of d , we use the chain rule

$$\frac{\partial \hat{f}}{\partial m} = \frac{\partial f}{\partial m} \cdot \frac{\partial \hat{f}}{\partial f}$$

$$\text{E.g. } \partial \log \rho^a / \partial \rho^a = 1 / \rho_a$$

Combined model and data transformation

$$\text{Sensitivity } S_{ij} = \frac{\partial \rho_i^a}{\partial \rho_j}$$

$$\text{Data } d_i = \log \rho_i^a, \text{ model parameter } m_i = \log \rho_j$$

⇒ Jacobian matrix

$$J_{ij} = \frac{\partial \log \rho_i^a}{\partial \log \rho_j} = \frac{\partial \rho_i^a}{\partial \rho_j} \cdot \frac{\rho_j}{\rho_i^a}$$

Gauss-Newton with regularization

$$\Phi = (\mathbf{d} - \mathbf{f}(\mathbf{m}))^T (\mathbf{d} - \mathbf{f}(\mathbf{m})) + \lambda (\mathbf{c} - \mathbf{C}\mathbf{m})^T (\mathbf{c} - \mathbf{C}\mathbf{m})$$

$$b_i = \frac{\partial \Phi}{\partial m_i} = -2\mathbf{S}^T (\mathbf{d} - \mathbf{f}(\mathbf{m})) - 2\lambda \mathbf{C}^T (\mathbf{c} - \mathbf{C}\mathbf{m})$$

$$B_{ij} = \frac{\partial^2 \Phi}{\partial m_i \partial m_j} = \frac{\partial b_i}{\partial m_j} \approx 2\mathbf{S}^T \mathbf{S} + 2\lambda \mathbf{C}^T \mathbf{C}$$

$$\Rightarrow (\mathbf{S}^T \mathbf{S} + \lambda \mathbf{C}^T \mathbf{C}) \Delta \mathbf{m}^n = \mathbf{S}^T (\mathbf{d} - \mathbf{f}(\mathbf{m}^n)) + \lambda \mathbf{C}^T (\mathbf{c} - \mathbf{C}\mathbf{m}^n)$$

Resolution matrices for non-linear problems

$$\mathbf{m}^{k+1} = \mathbf{m}^k + \mathbf{S}^\dagger(\mathbf{d} - \mathbf{f}(\mathbf{m})) + \mathbf{C}^\dagger \mathbf{C}(\mathbf{m}^0 - \mathbf{m}^k)$$

with

$$\mathbf{S}^\dagger = (\mathbf{S}^T \mathbf{S} + \lambda \mathbf{C}^T \mathbf{C})^{-1} \mathbf{S}^T$$

$$\mathbf{C}^\dagger = (\mathbf{S}^T \mathbf{S} + \lambda \mathbf{C}^T \mathbf{C})^{-1} \mathbf{C}^T$$

and their combination $\mathbf{S}^\dagger \mathbf{S} + \mathbf{C}^\dagger \mathbf{C} = \mathbf{I}$

Resolution matrices for non-linear problems

$$\mathbf{m}^{k+1} = \mathbf{m}^k + \mathbf{S}^\dagger(\mathbf{d} - \mathbf{f}(\mathbf{m})) + \mathbf{C}^\dagger \mathbf{C}(\mathbf{m}^0 - \mathbf{m}^k)$$

assuming we “come close”, i.e. $\mathbf{m}^{est} = \mathbf{m}^{k+1}$ and $\Delta \mathbf{m}^k$ is small. With

$$\mathbf{d} = \mathbf{f}(\mathbf{m}^{true}) + \mathbf{n} = \mathbf{f}(\mathbf{m}^k) + \mathbf{S}(\mathbf{m}^{true} - \mathbf{m}^k) + \mathbf{n}$$

$$\Rightarrow \mathbf{m}^{est} = \mathbf{m}^k + \mathbf{S}^\dagger \mathbf{S}(\mathbf{m}^{true} - \mathbf{m}^k) + \mathbf{C}^\dagger \mathbf{C}(\mathbf{m}^0 - \mathbf{m}^k) + \mathbf{S}^\dagger \mathbf{n}$$

$$\Rightarrow \mathbf{m}^{est} = \mathbf{m}^k + \mathbf{S}^\dagger \mathbf{S} \mathbf{m}^{true} - (\mathbf{S}^\dagger \mathbf{S} + \mathbf{C}^\dagger \mathbf{C}) \mathbf{m}^k + \mathbf{C}^\dagger \mathbf{C} \mathbf{m}^0 + \mathbf{S}^\dagger \mathbf{n}$$

$$\Rightarrow \mathbf{m}^{est} = \mathbf{R}^M \mathbf{m}^{true} + (\mathbf{I} - \mathbf{R}^M) \mathbf{m}^0 + \mathbf{S}^\dagger \mathbf{n}$$

Data resolution for non-linear problems

$$\mathbf{f}(\mathbf{m}^{k+1}) = \mathbf{S}(\mathbf{S}^\dagger(\mathbf{f}(\mathbf{m}^{true}) + \mathbf{d} - \mathbf{f}(\mathbf{m}^k))) + \mathbf{C}^\dagger \mathbf{C}(\mathbf{m}^0 - \mathbf{m}^k)$$

$$\mathbf{f}(\mathbf{m}^{est}) = \mathbf{R}^D \mathbf{f}(\mathbf{m}^{true}) + \mathbf{R}^D \mathbf{n} + (\mathbf{I} - \mathbf{R}^D) \mathbf{f}(\mathbf{m}^0)$$

$$\mathbf{f}(\mathbf{m}^{est}) = \mathbf{R}^D (\mathbf{f}(\mathbf{m}^{true}) + \mathbf{n}) + (\mathbf{I} - \mathbf{R}^D) \mathbf{f}(\mathbf{m}^0)$$

$$\mathbf{f}(\mathbf{m}^{est}) = \mathbf{R}^D \mathbf{d} + (\mathbf{I} - \mathbf{R}^D) \mathbf{f}(\mathbf{m}^0)$$

Response through data space & starting response through null space